

SOP-08: Sodium Hydroxide and Corrosive Bases

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Room 213
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	_____ Date: _____
This SOP was created by (if not PI)	_____ Date: _____
Emergency contact	9-1-1 (campus phone) EH&S 206-543-7262 (M-F 8-5) 206-685-UWPD (8973) after hours
Date of last review	_____ Next review due: _____ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
Sodium hydroxide, solid pellets (500 g, BP359)	1310-73-2
Sodium hydroxide solution (1 L, catalog 0338; two containers)	1310-73-2
pH 10.01 and pH 10.00 buffer solutions (dilute base; low hazard, included for completeness)	mixture

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Corrosion (skin corrosion Cat 1A, eye damage Cat 1, corrosive to metals); Exclamation mark

- Stock chemicals: sodium hydroxide pellets (500 g) and sodium hydroxide solution (1 L containers). Intermediates and working solutions: 1M, 0.1M and 10M NaOH prepared for pH adjustment of buffers and experimental seawater, for RNase decontamination of surfaces and equipment, and for glassware cleaning. Wastes: spent base solutions, base rinses, and neutralized aqueous waste where treatment is authorized.
- Sodium hydroxide is one of the most damaging chemicals in this inventory for eye injury. Alkali burns of the eye penetrate deeper and faster than acid burns and are a leading cause of chemical blindness. Eye protection is not optional at any scale of work.

- Routes of exposure: skin and eye contact (primary), inhalation of dust from pellets or of mist from concentrated solutions, ingestion.
- How exposure might occur: weighing pellets (which are hygroscopic and become slippery and caustic in humid air), dissolving pellets in water (strongly exothermic), pouring or pipetting concentrated solution, splash during pH adjustment, and contact with residue on the outside of containers.
- Target organs: eyes, skin, respiratory tract, gastrointestinal tract on ingestion.
- Signs and symptoms of exposure: a soapy or slippery feel on the skin, which indicates that the base is already saponifying skin lipids; redness, pain and deep burns; severe eye pain, corneal clouding and possible permanent vision loss; coughing and airway irritation from dust or mist. Injury may progress for hours after contact.
- Physical and reactivity hazards: sodium hydroxide is a Class 1 water-reactive material - dissolving pellets in water releases substantial heat and can cause the solution to boil and spatter. Add pellets slowly to a large volume of cold water with stirring, never water to pellets. Sodium hydroxide reacts violently with acids, attacks aluminum, zinc, tin and galvanized metal to release flammable hydrogen gas, and reacts with chloroform to form dichlorocarbene (see SOP-03).
- Expected by-products: hydrogen gas from base-metal contact; heat from neutralization and from dissolution.
- GHS classification (from MyChem / SDS): Corrosive to metals Category 1; Serious eye damage/eye irritation Category 1; Skin corrosion/irritation Category 1A; Specific target organ toxicity - single exposure Category 3 (respiratory irritation or drowsiness/dizziness). MyChem fire class: Water Reactive, Class 1; Corrosive.
- Occupational exposure limit: sodium hydroxide 2 mg/m³ ceiling (WISHA/OSHA). Sodium hydroxide is NOT flagged as a particularly hazardous substance in MyChem; PHS Sections 9-11 are optional and are included below as good practice.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- Weighing sodium hydroxide pellets and preparing solutions of 1M or greater is performed inside a certified chemical fume hood, or at minimum behind a bench shield with the container in a water-ice bath, to control dust and spatter. Any chemical fume hood used must be tested and passed by EH&S. The certified hood in FTR 209 is used when a hood is required; work originating in FTR 213 moves to FTR 209 for those steps unless a current certified hood is available in FTR 213.
- Dilute solutions (0.1M or less, and commercial pH 10 calibration buffers) may be handled at the bench.
- Solutions are prepared in heat-resistant borosilicate glass or in polypropylene, never in aluminum, zinc or galvanized containers.
- A base spill kit containing a neutralizing absorbent (citric acid based or a commercial caustic neutralizer with color indicator) is maintained in FTR 213 and FTR 209.
- The nearest eyewash and safety shower in each room are inspected and flushed weekly by lab staff. Because alkali eye burns worsen with every second of delay, the eyewash location is confirmed and the path kept clear before any base work begins; a blocked eyewash is a stop-work condition.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: buttoned barrier lab coat. Because sodium hydroxide is corrosive and causes deep burns, additional protective clothing is REQUIRED where splashes or skin contact are foreseeable: a chemically resistant (PVC or neoprene) apron plus a face shield for weighing pellets, for preparing solutions of 1M or greater, and for all spill cleanup. Disposable chemically resistant sleeves are worn for transfers of more than 100 mL of concentrated solution.
- Base-contaminated clothing is removed immediately, under the safety shower if the skin is involved.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Required gloves: butyl rubber or neoprene gloves over an inner 4-6 mil nitrile glove for handling pellets and for solutions of 1M or greater. Nitrile alone (6-8 mil) is acceptable for solutions below 1M and for pH buffer handling, changed immediately on any contact.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with the specific chemical being used. Gloves must be inspected prior to use, including a check for pinholes.
- Sodium hydroxide on a glove feels slippery; that sensation means the base is present and the glove must be removed and replaced immediately, and the hand washed and inspected.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with these chemicals. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 chemical splash goggles for ALL work with sodium hydroxide at any concentration above pH 10 buffer, including small transfers. A face shield worn over goggles is REQUIRED for weighing pellets, for preparing solutions of 1M or greater, and for all spill cleanup. Safety glasses with side shields are acceptable only for commercial pH 10 calibration buffer use.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP when they are performed as written.
- Sodium hydroxide solutions below 1M may be used outside of the fume hood at the bench. Weighing pellets and preparing solutions of 1M or greater must be done in a certified chemical fume hood or behind a bench shield with dust control as described in Section 3.
- No lab member is currently enrolled in the EH&S Respiratory Protection Program for this chemical.
- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit (AL) or Occupational Exposure Limit (OEL) has been exceeded or when there is a possibility that an AL/OEL will be exceeded. If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator to discuss respiratory protection or to enroll in the program so a respiratory protection analysis can be performed. Program enrollment includes medical evaluation, training and fit testing for an appropriate respirator.
- Where air-purifying respirators are appropriate, use a full-face respirator with an appropriate particulate/acid-gas combination cartridge as a backup to engineering controls. Use a full-face supplied air respirator if it is the sole means of protection.

Section 4 - Special Handling and Storage Requirements

- ALWAYS ADD BASE TO WATER, NEVER WATER TO BASE, and add pellets slowly to a large volume of COLD water with continuous stirring, using an ice-water bath for solutions of 1M or greater.
- Quantity limits: no more than 100 g of pellets is weighed in a single operation; no more than 1 L of concentrated (1M or greater) solution is prepared at one time.
- Keep the pellet bottle tightly closed. Sodium hydroxide pellets absorb water and carbon dioxide from the air, cake, and can weld the cap to the bottle; a bottle that will not open is not forced - contact the PI.
- Never mix sodium hydroxide with acid except during controlled, approved neutralization performed slowly with cooling. Never use sodium hydroxide with chloroform (SOP-03), with aluminum or zinc, or with ammonium salts.
- Use borosilicate glass or polypropylene labware only; sodium hydroxide etches ordinary glass over time and dissolves aluminum.
- All working containers and dilutions must be labeled with the full chemical name, the concentration, the word 'CORROSIVE', the date and the preparer's initials.
- Concentrated base work is performed only when at least one other person is present in or immediately adjacent to the room. Unattended operations with open concentrated base are prohibited; a covered NaOH glassware soak bath is permitted if posted with contents, owner, date and a hazard warning.
- A 'CORROSIVE BASE IN USE' sign is posted on the hood sash or bench while work is in progress.
- Storage: sodium hydroxide is stored below eye level in the corrosives cabinet in FTR 213 (and FTR 209 where present), in a secondary containment tray, physically separated from all acids, from metals, and from chloroform and other halogenated solvents. The base storage position is clearly distinguished from the acid position within the cabinet.
- Transport: bottles are carried in a bottle carrier; solutions are moved in closed secondary containers.
- Best practices to minimize accidents: pre-stage the balance, weigh boat, spatula and receiving vessel before opening the pellet bottle; keep neutralizer within reach; wipe container exteriors after use so no caustic residue is left for the next person.
- Clean the fume hood or work surface upon completion of tasks with clean water followed by a dilute citric acid or vinegar solution and then clean water; verify neutrality with pH paper and wipe dry.
- Clean all contaminated surfaces with water, then dilute weak acid, then water, and dry.
- Place all contaminated disposable items in appropriate laboratory waste for disposal.
- Non-disposable/re-usable utensils, glassware, and other surfaces contaminated with sodium hydroxide must be decontaminated at the end of the laboratory work session.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.
- Small liquid spill - less than 250 mL of solution below 1M at the bench, or less than 100 mL of concentrated solution inside the fume hood: may be handled by trained lab staff. Wearing goggles, face shield, butyl or neoprene gloves over nitrile, lab coat with apron, apply neutralizing absorbent from the base spill kit around the perimeter first and then inward, wait for the indicator to show neutralization is complete, and scoop the residue into a screw-top container with a plastic scoop.

- Small PELLET spill: do not sweep or brush pellets dry, which scatters caustic dust. Pick up pellets with a plastic scoop or with damp absorbent, working from the outside inward, and place them in a screw-top container. Then neutralize and wipe the residue as above.
- Any spill greater than 250 mL, any spill of concentrated solution outside the fume hood, any spill that reaches aluminum or other reactive metal, and any spill in which a person may have been exposed: DO NOT clean up. Evacuate the room, close the door, post a warning sign, prevent entry, and call 9-1-1 from a campus phone. Notify EH&S.
- PPE required for cleanup: chemical splash goggles AND face shield, butyl or neoprene gloves over 6-8 mil nitrile, buttoned lab coat with chemically resistant apron and disposable sleeves, full-length pants and close-toe shoes.
- Signage and entry restrictions: post 'DO NOT ENTER - CAUSTIC SPILL' on the door for any large spill; restrict entry until EH&S clears the space.
- Spill area must be cleaned up in the following manner: after neutralizer removal, verify neutrality with pH paper - sodium hydroxide leaves an invisible, slippery caustic film that must be neutralized and rinsed, not merely wiped - then clean the area thoroughly with detergent solution followed by clean water, and dry.
- Spill cleanup materials must be disposed of in the following manner: place neutralized residue and contaminated disposables in a double-bagged, screw-top container, label with a UW Hazardous Waste Label listing 'sodium hydroxide - neutralized spill debris', and submit a collection request to EH&S.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Eye exposure: use the eyewash IMMEDIATELY for at least 15 minutes - ideally 30 minutes for sodium hydroxide - while holding the eyelids open, then go immediately to the Emergency Department. Alkali eye burns penetrate rapidly and continue to damage tissue; length of irrigation is the single most important determinant of outcome. Do not stop to find someone to help.
- Skin exposure: use the nearest safety shower for at least 15 minutes; stay under the shower and remove all contaminated clothing including shoes; use a clean lab coat or spare clothing for cover-up. Continue flushing until the skin no longer feels slippery. Do not apply neutralizers or ointments. Seek medical evaluation for any burn.
- Inhalation exposure: move the person out of the contaminated area to fresh air; get medical help.
- Ingestion: do NOT induce vomiting. Rinse the mouth with water and give small sips of water if the person is conscious and able to swallow. Call 9-1-1 immediately.
- Sharps injury (needle stick or subcutaneous exposure): scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap; seek medical evaluation.
- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent and concentration, dose, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance. Secure the area before leaving; lock doors and indicate the spill if needed.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams: (a) unused sodium hydroxide pellets or solution; (b) spent base solutions and glassware soak baths; (c) base rinses; (d) base-contaminated solid debris; (e) neutralized aqueous waste where treatment is authorized.
- Keep base waste SEPARATE from acid waste, from halogenated solvent waste (chloroform in particular), from oxidizer waste, and from metal-containing waste.
- In-lab treatment: dilute sodium hydroxide solutions of pH 12.5 or below that contain no other hazardous constituent may be neutralized to pH 6-9 with dilute acid or citric acid and, if EH&S guidance for the specific solution permits, disposed of to the sanitary sewer with copious water. Neutralization is performed slowly, in a fume hood or behind a shield, with cooling, stirring and pH paper verification, wearing full corrosive PPE. Solutions above pH 12.5, and any base solution containing metals, solvents or other hazardous constituents, are collected as hazardous waste rather than treated.
- Accumulate liquid waste at the point of generation in a sturdy, chemically compatible polypropylene or HDPE container with a securely closable screw-top lid, kept closed except when actively adding waste, in secondary containment. Do not use glass for concentrated base (etching) or any metal container.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate percentage, starting from the moment the first drop of waste is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- Be aware that many laboratory accidents happen from inadvertent disposal of incompatible wastes into the same waste container - adding base waste to an acid waste container is a common and dangerous error. Therefore, identify different waste streams as appropriate and label them clearly.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste. Never autoclave chemical waste because it can produce hazardous chemical vapors, aerosols, and explosive reactions.
- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University, all its campuses, and off-site locations; University laboratories cannot contract with an outside vendor to collect hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials. Update the MyChem inventory when containers are emptied, received, transferred or surplus.

Section 7 - Protocol

- Procedure A - Preparation of sodium hydroxide solutions from pellets. In the certified fume hood or behind a bench shield, wearing goggles, face shield, butyl over nitrile gloves, lab coat and apron: place approximately 80% of the final volume of cold deionized water in a borosilicate or polypropylene vessel in an ice-water bath. Weigh the pellets into a plastic weigh boat on the balance (keep the pellet bottle capped between scoops) and add them slowly to the stirring water in small portions, allowing the solution to cool between additions. When fully dissolved and cooled to room temperature, bring to final volume, transfer to a labeled polypropylene bottle, and label with name, molarity, date and initials.
- Procedure B - pH adjustment of buffers and experimental seawater. Add dilute NaOH (0.1M or 1M) dropwise with stirring and continuous pH monitoring, at the bench, wearing nitrile gloves and splash goggles.
- Procedure C - RNase decontamination and glassware cleaning. Prepare 0.1M-0.5M NaOH and apply to surfaces with a wipe or soak glassware in a covered, posted bath. Rinse thoroughly with deionized water, collecting the first rinse into the base waste container, and verify neutrality with pH paper before returning items to service.
- Procedure D - pH meter calibration with pH 10 buffer. Handle commercial buffer at the bench with nitrile gloves and safety glasses; used buffer may be discarded to the drain with running water after confirming with the SDS.

- Prohibited: adding water to pellets; mixing base with acid outside a controlled neutralization; using NaOH with chloroform, aluminum, zinc or ammonium salts; storing base with acids; forcing a caked pellet bottle; use by untrained personnel.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use for particularly hazardous substances (PHSs).
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

No. N/A - sodium hydroxide is not administered to animals under this SOP. Dilute base is used only for pH adjustment of experimental seawater; if that seawater is used with live animals, the exposure is conducted under the applicable IACUC-approved protocol and the base handling still follows this SOP.

IACUC protocol number (if applicable): _____

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 9 - Approvals Required

- All staff working with sodium hydroxide must be trained on this SOP prior to starting work. They must also review the sodium hydroxide SDS, and it must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.
- Prerequisite training, all documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) EH&S Managing Laboratory Chemical Waste training; (3) documented review of this SOP and the SDS; (4) documented instruction on exothermic dissolution, acid-base incompatibility, and the location and use of the eyewash and safety shower.
- A worker must complete at least one supervised preparation of a 1M or greater solution (Procedure A), observed and signed off by the PI or a designated experienced lab member, before performing it independently.
- Neutralization of base waste for sewer disposal requires prior confirmation from the PI that the solution contains no other hazardous constituent, and adherence to EH&S chemical treatment guidance.
- No medical examination is required for the work described because respirators are not used. A medical examination and fit testing would be required before any respirator use.

Section 10 - Decontamination

- Work surface and fume hood: wipe with clean water, then dilute citric acid or white vinegar solution, then clean water; verify neutrality with pH paper and wipe dry. A surface that still feels slippery has not been decontaminated.
- Equipment: rinse pH probes per the manufacturer's instructions; rinse cylinders, beakers and stir bars into the base waste container, then wash with detergent and water and rinse three times with deionized water. Inspect any aluminum equipment near base work for corrosion.
- Glassware: rinse three times, collecting the first rinse as base waste, then wash with detergent and water and rinse three times with deionized water. Do not leave concentrated NaOH in glass volumetric ware, which it etches.
- Balance: brush any pellet fragments into a waste container with a plastic brush kept for this purpose, then wipe the pan with a damp wipe followed by a dry one. Do not leave caustic residue on a shared balance.
- Controlled areas: there are no glove boxes or restricted-access hoods in the Roberts Lab space.

- Cleaning solutions and materials: clean water, dilute citric acid or vinegar, pH paper, laboratory detergent solution, deionized water, lint-free wipes, and caustic-neutralizing absorbent.

Section 11 - Designated Area

- DESIGNATED AREA: pellet weighing and preparation of solutions of 1M or greater are performed in the certified chemical fume hood in FTR 209, or behind a bench shield at the designated corrosives bench in FTR 213 with the vessel in an ice-water bath.
- The designated area is posted with a 'DESIGNATED AREA - CORROSIVE BASE - SODIUM HYDROXIDE - EYE DAMAGE HAZARD' sign listing the PI, contact number and required PPE.
- Base waste containers are kept within the designated area in secondary containment, clearly separated from any acid waste container.
- Storage: the corrosives cabinet in FTR 213 (and FTR 209 where present), below eye level, in secondary containment, with base physically separated from all acids.
- Dilute solutions (0.1M or less) and commercial pH 10 buffers may be used at any laboratory bench in FTR 209 and FTR 213 with the PPE specified in Section 3.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.

I have read and understand the content of this SOP:

[illegible]

Name (print)	Signature	Date